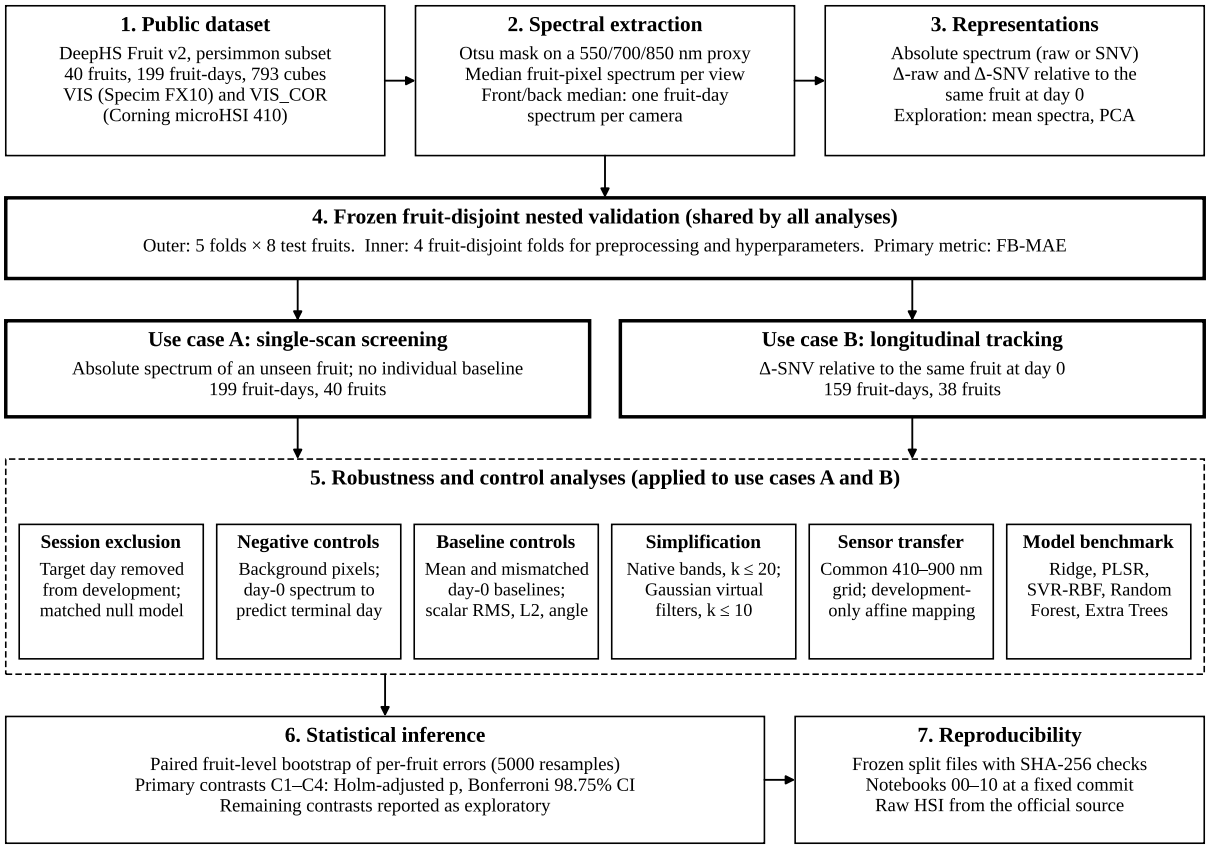
1. Public dataset
DeepHS Fruit v2, persimmon subset
40 fruits, 199 fruit-days, 793 cubes
VIS (Specim FX10) and VIS_COR
(Corning microHSI 410)
2. Spectral extraction
Otsu mask on a 550/700/850 nm proxy
Median fruit-pixel spectrum per view
Front/back median: one fruit-day
spectrum per camera
3. Representations
Absolute spectrum (raw or SNV)
Δ-raw and Δ-SNV relative to the
same fruit at day 0
Exploration: mean spectra, PCA
4. Frozen fruit-disjoint nested validation (shared by all analyses)
Outer: 5 folds × 8 test fruits. Inner: 4 fruit-disjoint folds for preprocessing and hyperparameters. Primary metric: FB-MAE
Use case A: single-scan screening
Absolute spectrum of an unseen fruit; no individual baseline
199 fruit-days, 40 fruits
Use case B: longitudinal tracking
Δ-SNV relative to the same fruit at day 0
159 fruit-days, 38 fruits
5. Robustness and control analyses (applied to use cases A and B)
Session exclusion
Target day removed from development; matched null model
Negative controls
Background pixels; day-0 spectrum to predict terminal day
Baseline controls
Mean and mismatched day-0 baselines; scalar RMS, L2, angle
Simplification
Native bands, k ≤ 20; Gaussian virtual filters, k ≤ 10
Sensor transfer
Common 410–900 nm grid; development-only affine mapping
Model benchmark
Ridge, PLSR, SVR-RBF, Random Forest, Extra Trees
6. Statistical inference
Paired fruit-level bootstrap of per-fruit errors (5000 resamples)
Primary contrasts C1–C4: Holm-adjusted p, Bonferroni 98.75% CI
Remaining contrasts reported as exploratory
7. Reproducibility
Frozen split files with SHA-256 checks
Notebooks 00–10 at a fixed commit
Raw HSI from the official source